ALEENA SAJI

Software Engineer
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PROFESSIONAL SUMMARY

B.Tech Computer Science and Engineering graduate and fresher Software Engineer with strong Flutter and Dart mobile application development skills. Independently built **Double Bell**, a Flutter and Firebase Android application currently in Google Play closed testing, using Firebase Authentication, Cloud Firestore, Riverpod state management and the Google Play Console release workflow. Solid software engineering fundamentals in Java, Python, SQL, REST/API concepts, data structures and debugging, with supporting experience in full-stack web development, machine learning and generative AI applications.

TECHNICAL SKILLS

Mobile Development: Flutter, Dart, Android Development, Android Studio, Mobile Application Architecture, Responsive & Adaptive UI, UI Development, Animation

State Management & App Development: Riverpod, Provider, GoRouter, Shared Preferences, Lottie

Backend, APIs & Databases: Firebase, Firebase Authentication, Cloud Firestore, REST APIs, JSON, SQL, PostgreSQL, MongoDB, Node.js, Express.js

Programming Languages: Dart, Java, Python, JavaScript, C, C#

Software Engineering: Git, GitHub, Debugging, Testing, Data Structures, Problem Solving, Design Patterns, Google Play Console, Visual Studio Code

Web Development: HTML, CSS, Bootstrap, JavaScript, PHP, Full-Stack Web Development

AI / ML / Data: Machine Learning, Deep Learning, Computer Vision, Generative AI, Data Science, SVM, Random Forest, Streamlit

PROJECT EXPERIENCE

Double Bell – Flutter Android Application — **Google Play Closed Testing**

Flutter, Dart, Riverpod, Firebase Authentication, Cloud Firestore, Google Sign-In, GoRouter, Git/GitHub

- Independently developed an end-to-end **Flutter/Dart Android application** for private bus travel, enabling schedule search by start location, destination, and arrival time; currently in **Google Play closed testing**.
- Integrated **Firebase Authentication, Google Sign-In, and Cloud Firestore** for user accounts, bus stop/schedule data, profile management, password changes, logout, and account deletion.
- Implemented **Riverpod** state management, **GoRouter** navigation, **Shared Preferences** persistence, responsive UI, **Lottie** animations, custom splash screen, **audioplayers**, and **URL launching**.
- Built time-based bus search with schedule display, **bus type information, incorrect-timing reporting, language selection**, and About, Privacy Policy, and Terms screens.
- Configured **Firebase production environment and Firestore security rules**, resolved application issues, and managed **Google Play closed testing** for production release preparation.

MammoSafe – Breast Cancer Detection Using Ultrasound, PET/CT and Histopathological Images

B.Tech Group Project | Python, Machine Learning, Deep Learning, Computer Vision, EfficientNet, SVM

- Developed the **histopathological image analysis module** as part of a 4-member AI-based breast cancer detection project using **Python**.
- Utilized **EfficientNet** for deep feature extraction from histopathological images and extracted discriminative features for accurate classification.
- Implemented **Support Vector Machine (SVM)** for classification and performed **image preprocessing, model training, and evaluation** for the histopathology component.
- Contributed to integrating histopathology predictions with **ultrasound and PET/CT models** using a **multi-modal fusion** approach to combine complementary medical imaging information.
- Collaborated with team members to integrate the machine learning models into a **Streamlit-based web application** supporting image upload, real-time prediction, and report generation.

AI-Powered Health & Nutrition Analysis App

Python, Streamlit, Google Gemini Vision API, Generative AI, PIL, python-dotenv

- Built a Streamlit application that uses the Gemini Vision API for multimodal analysis of food images, identifying food items and estimating calories, protein, carbohydrates and fats for each.
- Implemented image upload, custom user prompts and AI-generated nutritional reports; processed images with PIL, managed API keys with python-dotenv and organized code into reusable functions.

Multi-Language Invoice Extractor

Python, Streamlit, Google Gemini 1.5 Flash API, Generative AI, PIL, python-dotenv

- Developed a Streamlit application that uses Gemini 1.5 Flash to process invoice images (JPG, JPEG, PNG) and answer natural-language queries about their content.
- Built an interface with invoice upload, image preview and custom prompts; handled images with PIL and secured API credentials using environment variables.

EDUCATION

Bachelor of Technology (B.Tech) – Computer Science and Engineering **2022 – 2026**
University College of Engineering, Thodupuzha (APJ Abdul Kalam Technological University) | CGPA: 8.25

Higher Secondary Education
St. Mary's HSS, Arakkulam | Score: 96.33%

SSLC
Sacred Heart English Medium Higher Secondary School, Moolamattom | Result: Full A+

CERTIFICATIONS

- **Flutter & Dart – The Complete Guide** – Udemy (Academind / Maximilian Schwarzmüller)
- **The Complete Full-Stack Web Development Bootcamp** – Udemy (Dr. Angela Yu)
- **Artificial Intelligence using Python** – Cozmek Private Limited, Kothamangalam (recognized as one of the best students)
- **Python for Data Science** – NPTEL
- **Foundational C# with Microsoft** – freeCodeCamp
- **Brototype Campus Community** – Completion Certificate

LEADERSHIP & ACTIVITIES

- **Robotics Club, University College of Engineering, Thodupuzha** – Member for 4 years; Core Member for the final 1.5 years. Took part in technical discussions, shared ideas and volunteered for programs, events and games.
- **Python Classes for B.Tech Students** – Conducted Python classes, building communication, mentoring and technical knowledge; recognized with a reward for teaching.
- **Election Grievances Redressal Cell** – Member; handled and resolved election complaints.

LANGUAGES

English, Malayalam, Hindi, Russian

REFERENCE

Mrs. Rahumath Beevi – HoD, Department of Computer Science and Engineering, University College of Engineering, Thodupuzha | Contact: +919207557512